

**Srini Chinamilli**

Co-Founder & CEO, Tessolve

AI chips and Indian innovation

Insights from Tessolve CEO **Srini Chinamilli****Jayesh Shinde** | jayesh@digit.in

When it comes to semiconductor chip companies, it's hard to look beyond the obvious choices like Intel, AMD, TSMC, NVIDIA, Qualcomm, MediaTek and Samsung. But ask any of these big chip companies, and they'll be the first ones to tell you that they are supported by an army of partners in bringing all the pieces of the semiconductor chip jigsaw puzzle together. Tessolve is one such important player in not just India's but the global chip design and manufacturing industry.

According to the Co-Founder and CEO, Srini Chinamilli, Tessolve's expertise lies in semiconductor and embedded technologies. "From smartphones and wearables to critical systems in automotive, industrial, and avionics sectors, our services are integral to the semiconductors powering these innovations," he says.

Founded in 2004, Tessolve is a multinational semiconductor engineering solution provider headquartered in Bangalore, India. The company serves a diverse customer base, including leading semiconductor manufacturers and technology companies. "We have designed parts for an interesting touch-sense application for SigmaSense, made SOMs with Renesas parts for industrial robots, systems for 2-wheeler dashboards, and even IOT devices for Kaapi

Machines," explains Chinamilli, to showcase some of the work they've been doing recently, among other things. Tessolve's had a hand in designing components for everything from industrial robots to IoT devices for coffee machines. But what truly sets them apart is their end-to-end approach – covering chip design, testing, product engineering, and system design – all under one roof.

As AI becomes more woven into the fabric of our everyday lives, I wanted to delve into how Tessolve is navigating the complex landscape of AI chip development. And perhaps, most intriguingly, how does Tessolve see India's role in the global semiconductor arena, especially as the country positions itself as a burgeoning hub for tech innovation? I posed these questions and more to Srini. His insights offer a fascinating glimpse into the future of technology and the pivotal role companies like Tessolve will play in shaping it. Edited excerpts follow:

Q What is it about semiconductor chip manufacturing that most people don't understand?

A As an intricate and highly specialised process that many people may not fully appreciate, semiconductor chip manufacturing involves multiple testing stages, validation, optimisation, and

reliability to ensure performance. The importance of these steps are to ensure proper performance, to the point where a minor defect can lead to significant issues in the final product, across different conditions is often overlooked.

At Tessolve, while we do not manufacture chips, our critical role comes in with the comprehensive design, testing, and validation of the process, with our expertise spanning initial chip design to post-silicon validation to ensure that each chip meets the highest quality and reliability standards prior to market-entry to meet the demands of industries like AI, automotive, and telecommunications, where reliability is crucial. To maintain these standards, we use advanced testing platforms capable of simulating real-world conditions, allowing us to detect and address potential issues early in the development cycle. Our engineers work closely with chip manufacturers to optimise designs for performance, energy efficiency, and scalability.

Q What differentiates Tessolve from other players in the AI chip landscape?

A Tessolve has evolved significantly since its establishment in 2004. We initially identified a gap in India's semiconductor testing and productisation capabilities, which motivated us to build